

Metal Contamination of Cracking Catalysts. 1. Synthetic Metals Deposition on Fresh Catalysts

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Metal contamination of cracking catalysts is a problem of growing concern due to the increasing interest in residual feedstocks for FCC units. In the past, investigations of the effects of metal contaminants were limited to lengthy pilot plant studies or difficult and inaccurate bench-scale studies. These problems have been overcome by the combined use of a microactivity test unit and a new synthetic method of contaminating fresh cracking catalysts. Several different techniques for synthetic contamination were investigated by comparing catalytic activities of fresh contaminated catalysts to the activities of equilibrium catalysts containing comparable metal levels. Excellent agreement with the equilibrium catalysts was obtained by heat shocking fresh catalysts at 1100 °F, impregnating nickel and vanadium to the point of incipient wetness with metal naphthenates in a benzene solution, and finally, steam purging the catalysts at 1350 °F. With this technique, a rapid yet accurate method of evaluating the metals tolerance of cracking catalysts is now available.

Introduction

In the catalytic cracking of gas oils and other feedstocks, metal contaminants such as nickel, vanadium, iron, copper, etc. are continuously deposited upon the catalyst surface. Due to their ability to catalyze dehydrogenation reactions, these deposited metals soon alter the product distribution obtained from FCC units if they are allowed to reach certain maximum levels. As the level of metal contaminants increases, the yields of carbon and hydrogen increase at the expense of gasoline production.

Presently, it is common practice for refineries to routinely add fresh catalyst and withdraw equilibrium catalyst from FCC inventories to maintain activity and prevent high metal levels on the catalyst. Due to a continually increasing demand for gasoline coupled with shrinking supplies of normally used cracking stocks, more attention is now being directed to the catalytic cracking of heavier charge stocks such as residuals. One of the major problems with these heavier stocks is that they contain considerably higher levels of metal contaminants and require such high catalyst make-up rates that processing them is often uneconomical with present catalysts. Highly contaminated gas oils, such as derived from California crudes, typically contain 1 ppm of nickel which is considered near the upper limit for today's catalysts. In contrast, California residuals contain over 100 ppm of nickel and would obviously deposit metal contaminants at a very rapid rate. Zeolitic cracking catalysts currently available commercially are more tolerant of metal contaminants than the older clay or amorphous catalysts, but they are still inadequate for residual processing. Also, at the present time, there is no economical means of removing the metal contaminants from the catalyst. Thus, there is an urgent need for an improved cracking catalyst that can tolerate high levels of metal contaminants.

Numerous articles and patents have dealt with the problem of metal contamination of nonzeolitic cracking catalysts (McEvoy et al., 1957; Mills, 1950; Meisenheimer, 1962; Foster et al., 1963; Adams and Sterba, 1963; Dilliplane et al., 1963; Leum and Conner, 1962). Past studies have also dealt with the effect of various metals upon activity (Mills, 1950; Duffy and Hart, 1952; McIntosh, 1954), mechanistic studies (McEvoy et al., 1957; Meisenheimer, 1962), and methods of removing metals from contaminated catalysts (Foster et al., 1963; Dilliplane et al., 1963).

In several articles a method of synthetically contaminating fresh nonzeolitic catalysts was discussed (Rothrock et al., 1957; Conner et al., 1957; Grane, et al., 1961). In these studies, metal naphthenates were dissolved in either furnace oil or gas oil and then passed over a fluidized bed of the fresh catalyst at elevated temperatures. This technique in itself was inadequate for reproducing actual equilibrium catalysts, since the metals were more active than naturally deposited metal contaminants. A deactivation of the synthetically deposited metals was achieved by treating the final catalyst alternately with oil or hydrogen and air to simulate the environment of circulating equilibrium catalyst. After approximately 20 cycles of this treatment at 900 °F, the activity of the catalyst lined out at levels similar to those of naturally contaminated catalysts. Thus, even though good correlation is obtained, this method is time consuming and requires large catalyst samples. More recently, this technique has been applied to zeolitic cracking catalysts (Cimbalo et al., 1972).

The need for a more metals tolerant catalyst is well established, but development of such a catalyst requires extensive studies of the type that only bench-scale equipment can provide. In this manner, one may more quickly gain a better understanding of the poisoning mechanism and the role of various catalyst components. Pilot-plant tests, though certainly useful for final evaluations, are handicapped by their lengthy nature. These shortcomings of former test procedures have now been overcome by the combined use of a microactivity test unit and a new synthetic method of contaminating fresh cracking catalysts. Realistic results have been obtained from synthetically contaminated catalysts which are in excellent agreement with equilibrium catalysts. This study has been confined to an evaluation of nickel and vanadium metals, since these are the major contaminants in FCC feedstocks.

Apparatus and Procedure

A modified microactivity test unit (MAT) was used to evaluate all catalysts. This unit is similar to the original MAT unit (Henderson and Ciapetta, 1967) except that feed is injected through a hypodermic needle permanently mounted in the reactor head and the liquid product recovery is established by weighing instead of volumetric measurements. A Kuwait gas oil with a boiling range of 500 to 800 °F was used as feed. The catalyst charge was 2.5 g of 10-20 mesh sized granules. All catalysts were

Table I. Inspections of Standard Samples of Catalyst Y

	catalyst					
	Y _A	Y _B	Y _C	Y _D	Y _E	Y _F
history	fresh	equilibrium	equilibrium	equilibrium	equilibrium	equilibrium
pretreatment ^a	heat shocked and steamed	calcined	calcined	calcined	calcined	calcined
chemical composition, wt %						
Ni	-	0.020	0.045	0.073	0.129	0.170
V	-	0.030	0.075	0.080	0.129	0.106
Fe	-	0.34	0.33	0.51	-	0.34
Na	0.44	0.56	0.47	-	-	0.48
Cu	-	0.018	0.019	0.022	-	-
C	-	0.019	0.003	-	-	-
nickel equivalent, ppm ^b	0	260	600	890	1548	1912
surface area, m ² /g	102	84.9	87.0	89.7	-	87.2

^a Heat shock at 1100 °F, steam purged at 1350 °F, and calcination at 1000 °F. ^b Nickel equivalent = ppm Ni + 1/5 ppm V.

formed identically by pressing 10 cm³ of powder in a 1-in. die at 8000 psi and then broken and sized by passing through sieves. In this manner uniformity of catalysts was ensured and skin effects were minimized. Other reaction conditions were 900 °F, 80 s catalyst contact time, 14 WHSV, and a total feed throughput of 1 cm³. Prior to testing but after forming, all fresh catalysts were pretreated to simulate equilibrium catalysts. This pretreatment consisted of a heat shock treatment at 1100 or 1200 °F for 1 h followed by a 10-h steam purge at 1350 °F at atmospheric pressure with 100% steam. Equilibrium catalysts were calcined 10 h at 1000 °F prior to evaluation.

A series of regenerated equilibrium catalyst samples were obtained from commercial FCC units and pilot plant units. These were samples of the same commercial FCC catalyst, designated catalyst Y, with the only difference being the amounts of metal contaminants on the catalyst.

Fresh samples of catalyst Y were contaminated synthetically by three different techniques. In one, nickel nitrate and vanadium oxalate salts were dissolved in water and impregnated on the catalysts to the point of incipient wetness. These samples were then treated in a variety of ways which will be discussed later. In a second method, metal naphthenates were used to impregnate nickel and vanadium from a benzene solvent. The naphthenates are derived from a mixture of naphthenic acids which are monocarboxylic, monocyclic, and completely saturated hydrocarbons. The metal content of the nickel naphthenate was 6 wt % and the vanadium naphthenate was 3 wt %. Thirdly, the metals were deposited by dissolving the naphthenates in kerosene and passing this solution over a fluidized bed of the sample for 1 h at 650 °F. For the most part, the metal contaminants were synthetically deposited after the heat shock treatment, but before the steam treatment. A few samples were prepared by metal deposition after the steam treatment.

The amount of metal contaminants on the catalyst is expressed in terms of nickel equivalent which is defined as the nickel content in ppm plus one-fifth the vanadium content in ppm. This definition arises from the fact that vanadium as a metal contaminant has approximately one-fifth the effect of nickel as a dehydrogenating agent (Hildebrand et al., 1973).

Results and Discussion

With the present concern over high metals FCC content feedstocks and the advent of commercial residual cracking, it is desirable to establish the level of metal contaminants which results in deactivation of a typical zeolitic commercial catalyst. In addition, a standard set of data for comparison was needed before various methods of synthetically contaminating fresh cracking catalysts could be investigated. For these purposes, a series of equilibrium,

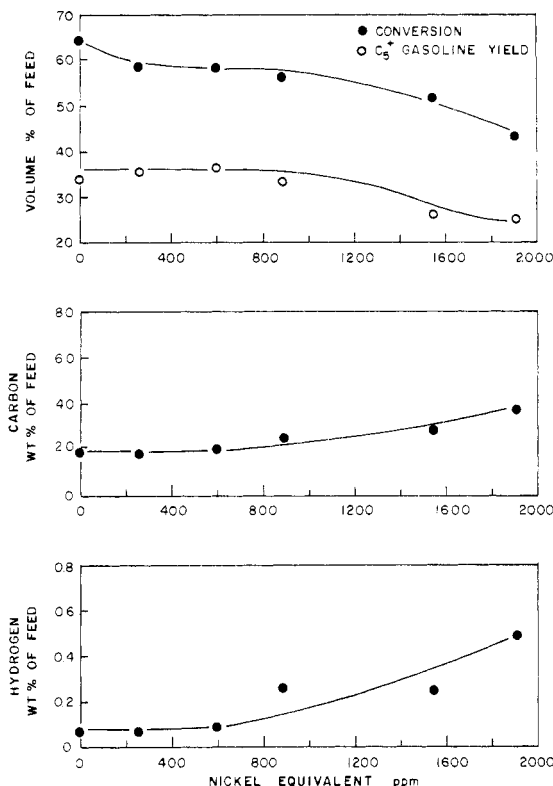


Figure 1. Effect of Ni and V Upon cracking characteristics of an equilibrium zeolitic catalyst.

naturally contaminated catalyst samples were obtained which contained varying amounts of nickel and vanadium contaminants. These were all samples of the same commercial zeolitic catalyst, designated catalyst Y. The descriptions and inspections of each equilibrium sample are shown in Table I, column Y_B through Y_F. In addition to the equilibrium catalysts, a fresh sample of catalyst Y was heat shocked and steam purged to simulate an equilibrium catalyst without metal contaminants; this is designated catalyst Y_A in Table I. The equilibrium samples range from 260 to 1912 ppm of nickel equivalent. However, surface areas and iron, sodium, copper, and carbon levels vary only slightly within the series except for the surface area of the fresh, steamed sample which is noticeably higher. Apparently, a more severe pretreatment of fresh samples is necessary to more effectively simulate an equilibrium catalyst.

The microactivities of these samples are shown in Figure 1 where conversion, gasoline yield, coke production, and hydrogen production are plotted as a function of nickel equivalent. As can be seen, the catalyst's activity is fairly

Table II. Metal Levels of Synthetically Contaminated Fresh Cracking Catalysts

nickel, wt %	vanadium, wt %	nickel equiv, ppm
0.020	0.030	260
0.045	0.075	600
0.060	0.125	850
0.075	0.225	1200
0.090	0.300	1500
0.170	0.106	1912

constant up to approximately 900 ppm of nickel equivalent. At higher metal levels, coke and hydrogen productions increase while conversion and C_5^+ gasoline yield decrease. The absence of sharp breaks in the curves makes it difficult to determine the exact point at which the metals first take effect, but it seems safe to conclude that the upper limit of metals tolerance for this particular catalyst is approximately 1000 ppm of nickel equivalent. Additional studies have shown that this is generally the upper limit for most of the zeolitic cracking catalysts in use today. At metal levels above 1000 ppm of nickel equivalent, metal deactivation becomes quite pronounced. These conclusions are in agreement with observations made during commercial operation. The higher conversion observed with fresh catalyst is due to its higher surface area relative to the equilibrium samples as discussed earlier.

Development of a residual cracking catalyst or a more metal tolerant catalyst requires extensive investigations. Pilot plant evaluations require large catalyst inventories, and appropriate metal levels are obtained only after lengthy on-stream periods. Accelerated metal deposition methods have been used, but even these are extremely time consuming (Cimbalo et al., 1972). Obviously, a reliable, yet rapid, means of contaminating and evaluating fresh cracking catalysts would expedite these studies. Several synthetic approaches have been examined, and the results are compared to the equilibrium samples shown in Figure 1. The levels of nickel and vanadium used for this study are shown in Table II.

In one approach a large batch of catalyst Y was heat shocked at 1200 °F (severe treatment) and steam purged prior to metal deposition. Samples from this batch were impregnated with aqueous solutions of nickel and vanadium salts and then given various post-treatments. The results, shown in Figures 2 and 3, are compared to the standard catalyst samples discussed earlier. In these and later figures, the data points have been omitted for clarity.

The critical nature of calcination temperature after metal deposition is shown in Figure 2. With a post-treatment temperature of 1000 °F, the metals are considerably more active on the synthetic samples than on the equilibrium samples. A more severe temperature treatment (1350 or 1500 °F) deactivates the metals to some extent and the data more closely approximate those of the standard catalyst samples. Agreement is still, however, far from acceptable even with the most severe temperature treatment. Heat treatments above 1500 °F were not attempted since structural collapse of the zeolite would probably have occurred.

Two synthetically contaminated samples which had been given a 1000 °F post-treatment were also treated alternately with hydrogen and oxygen at 900 °F. The purpose of this treatment was to simulate the environment commercial catalysts are exposed to during processing. In Figure 3 the activities of these samples are compared to the standard catalysts and the synthetic samples without the hydrogen-oxygen treatment. As can be seen, the cracking characteristics of the hydrogen-oxygen-treated

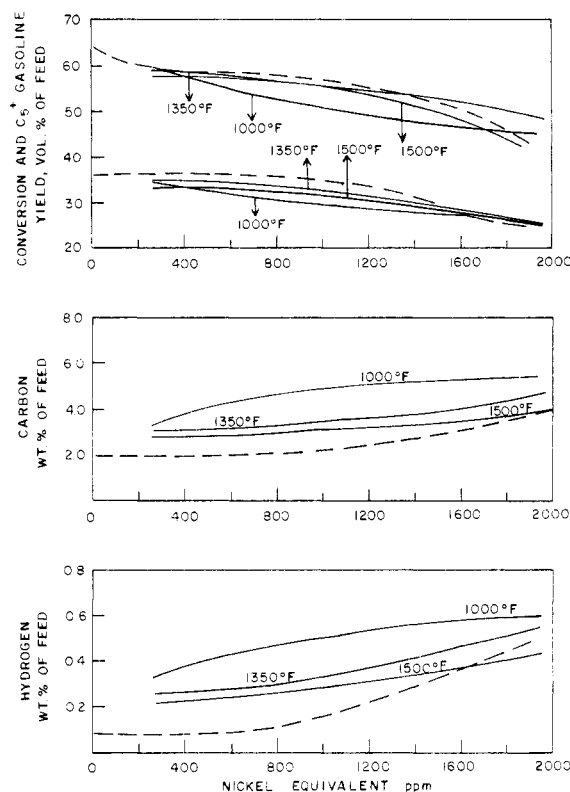


Figure 2. Metal salts impregnated on severely heat shocked, steamed samples of fresh catalyst Y: ---, equilibrium samples; —, synthetically contaminated samples. Temperatures refer to final calcination temperature.

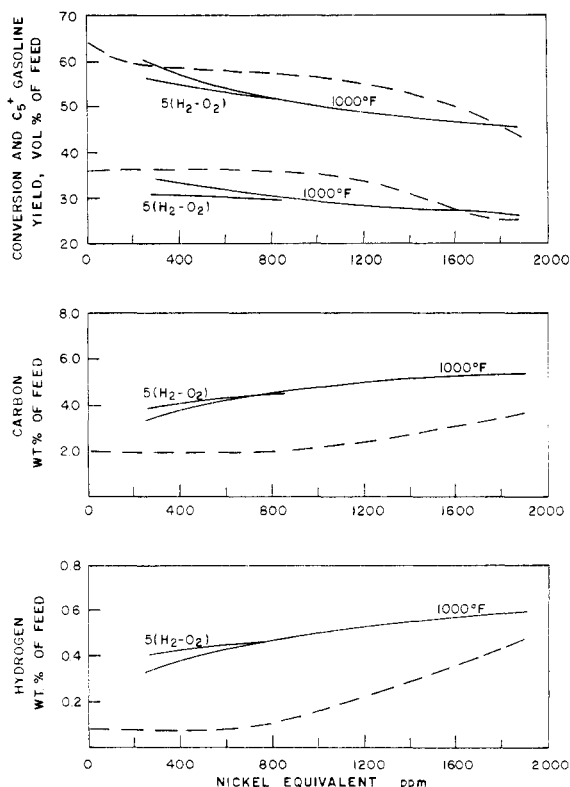


Figure 3. Effect of hydrogen-oxygen cyclic treatments at 900 °F: ---, equilibrium samples; —, synthetically contaminated samples.

samples were essentially the same as the untreated catalyst and, therefore, in poor agreement with the equilibrium catalysts.

Thus, it appears that fresh cracking catalysts which have been heat shocked and steam purged prior to metal im-

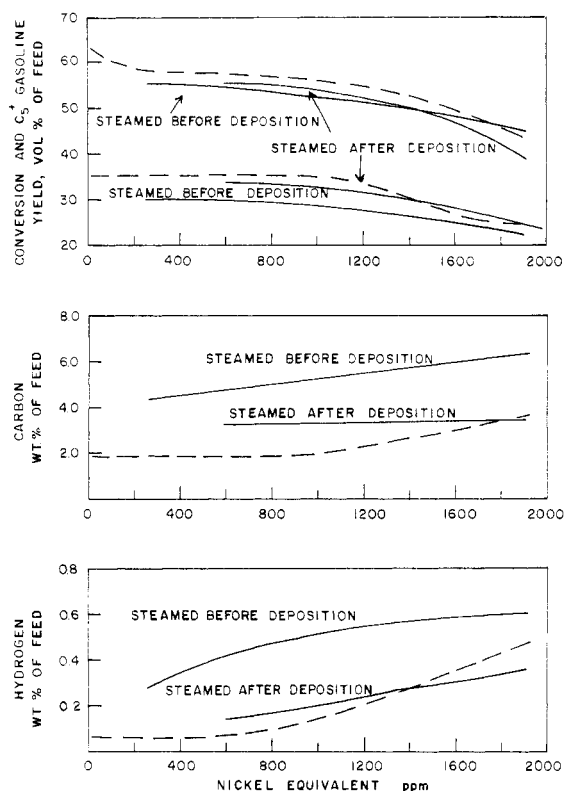


Figure 4. Effect of steaming before and after synthetic metal deposition: ---, equilibrium samples; —, synthetically contaminated samples.

pregnation do not give realistic cracking patterns regardless of the final treatment. The impregnated metals are overly active and produce more coke and hydrogen than observed with naturally contaminated catalyst samples.

The effect of steaming after synthetic metal deposition is shown in Figure 4. A less severe heat shock treatment (1100 °F) was used for these samples. Because of this variation, data for samples pretreated identically, except that the steam treatment was done prior to metals deposition, are included in Figure 4 for comparison. The results show conclusively that the order of metal impregnation and steam purging is a critical factor. The cracking characteristics of the synthetically contaminated fresh catalysts were very similar to the equilibrium samples when the fresh catalyst was treated in the following manner: (1) heat shock at 1100 °F; (2) impregnation of nickel and vanadium salts; (3) calcination at 1000 °F; (4) steam purge at 1350 °F. It has already been shown that the final calcination treatment does not properly deactivate the metals. Therefore, the key to realistically synthesizing metal contaminated cracking catalysts lies in treating the catalyst with steam after metals deposition.

It is obvious from Figures 2 and 3 that the metals deposited on catalysts in situ during commercial processing are not as active as comparable metal levels impregnated on the fresh catalysts. It is felt that the severe conditions encountered in cyclic FCC units cause partial deactivation of the metal contaminants. In addition, the steam treatment of fresh, impregnated catalysts deactivates the fresh metals in a similar manner resulting in very good agreement with the activity of the metal contaminated equilibrium catalyst. The mechanism by which metal deactivation occurs both in commercial operation and by laboratory steam treatment is not known, but a number of processes could account for this observation, namely: (1) slight structural collapse to mask metal contaminants; (2) formation of mixed metal oxides such as nickel aluminate

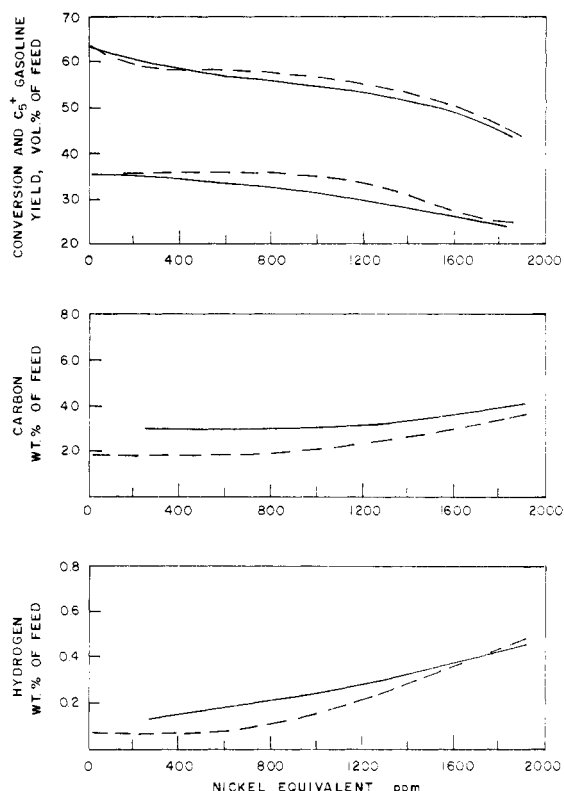


Figure 5. Metal naphthenates impregnated on fresh catalyst Y before steam treatment: ---, equilibrium catalysts; —, synthetically contaminated samples.

or silicate which may be less active; (3) migration and agglomeration of metal crystallites with reduction in effective metal surface area; (4) migration of metal to less active sites such as found on the matrix material. The occurrence of any one or combination of the above during high-temperature processing or steaming could result in decreased activity of the freshly deposited metal contaminants.

Though fairly good agreement was obtained with metal deposition before steam purging, the use of metal salts in water as a means of metal impregnation is not very realistic. Thus, several samples of catalyst Y were contaminated by impregnation with benzene solutions of nickel and vanadium naphthenates. Prior to evaluation in the MAT, these catalysts were steam purged. The cracking characteristics of samples contaminated by this technique are compared to those of the equilibrium catalyst samples in Figure 5. The agreement is quite satisfactory and is obviously better than all other synthetic methods of metal deposition. A few samples were impregnated with metal naphthenates after steaming, and the results were poor, which again emphasizes the importance of steam treating after metal deposition.

Finally, a synthetic method was evaluated which more closely simulates metal deposition during commercial operation than any other method in this study. Appropriate amounts of the metal naphthenates were dissolved in kerosene and the solution passed over a fluidized bed of fresh catalyst Y at 650 °F. This technique is similar to one reported earlier (Cimbalo et al., 1972) except that the final alternating treatments with hydrogen and oxygen were omitted. Samples were steamed either before or after the synthetic metal deposition. A comparison of the standard equilibrium catalyst samples and the fresh samples steamed after synthetic metal contamination is shown in Figure 6. Contrary to expectations, the metal effect with catalysts contaminated by this technique was much more

Table III. Description and Rating of Synthetic Methods Evaluated

source of metals	solvent	treatment before deposition	type of deposition	treatment after deposition	rating
metal salts	water	severe HS-ST ^a	impregnation	calcined at 1000 °F	very poor
metal salts	water	severe HS-ST	impregnation	calcined at 1350 °F	poor
metal salts	water	severe HS-ST	impregnation	calcined at 1500 °F	fair
metal salts	water	severe HS-ST	impregnation	5 cycles H ₂ -O ₂ at 900 °F	very poor
metal salts	water	mild HS-ST	impregnation	calcined at 1000 °F	very poor
metal salts	water	mild HS	impregnation	ST	good
metal naphthenates	benzene	mild HS-ST	impregnation	calcined at 1000 °F	poor-fair
metal naphthenates	benzene	mild HS	impregnation	ST	very good
metal naphthenates	kerosene	mild HS-ST	fluidized bed	calcined at 1000 °F	poor
metal naphthenates	kerosene	HS	fluidized bed	ST	poor

^a HS = heat shock; ST = steam purge.

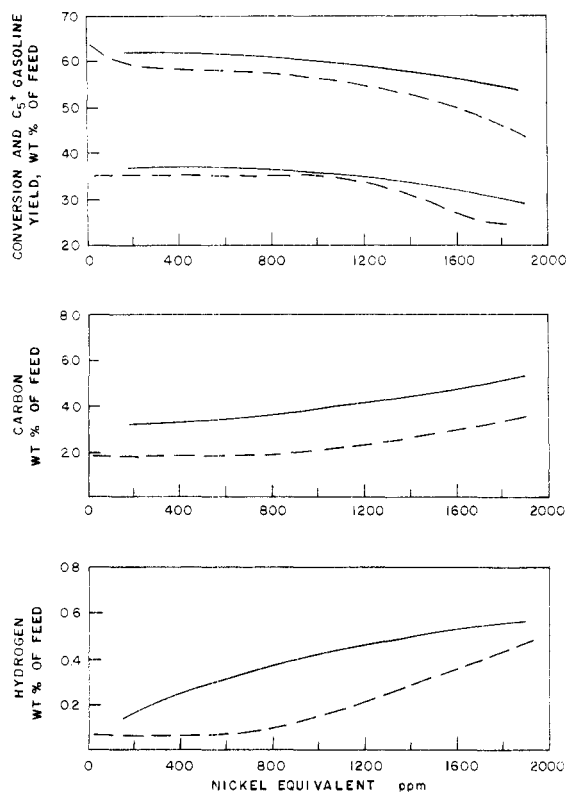


Figure 6. Metal naphthenates deposited on fresh catalyst Y in a fluidized bed at 650 °F: ---, equilibrium samples; —, synthetically contaminated samples.

pronounced than the equilibrium catalysts and the impregnated catalysts. As expected, samples steamed prior to the fluidized deposition showed even poorer agreement.

Conclusions

A description of the techniques evaluated for synthetic metal contamination and the results are summarized in Table III. The most critical factor in synthetic metal contamination of fresh catalysts is a post-steam treatment. Thus, after synthetic metal deposition by impregnation, the fresh catalyst must be steam purged at 1350 °F to deactivate the metals to levels comparable to naturally

contaminated equilibrium catalysts. It was also found that the most realistic results were obtained using nickel and vanadium naphthenates in a benzene solvent for impregnation of the metal contaminants.

Because of the synthetic technique developed in this study, it is now possible to rapidly and accurately evaluate the metals tolerances of large numbers of cracking catalysts without relying solely upon lengthy pilot plant runs. In addition, only small samples of catalysts are required since a microactivity test is used to obtain cracking activities. Several pilot plant aging runs have confirmed the validity of this technique. The rates of catalyst deactivation with synthetic contamination and MAT evaluation are in good agreement with pilot plant aging runs using identical catalysts. In addition, it has been found that a ranking of several catalysts for metal tolerances by the synthetic technique is in excellent agreement with pilot plant data.

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